

OMRON SYSMAC protocol MODEM driver

Overview

The Adroit Omron protocol driver provides serial Modem connectivity using RS-232 to omron host link modules The C200 and C1000 ,CS1 series amongst others, as well as KEYENCE PLC and Process-Plus modules are supported in this driver. Other PLCs may be compatible.

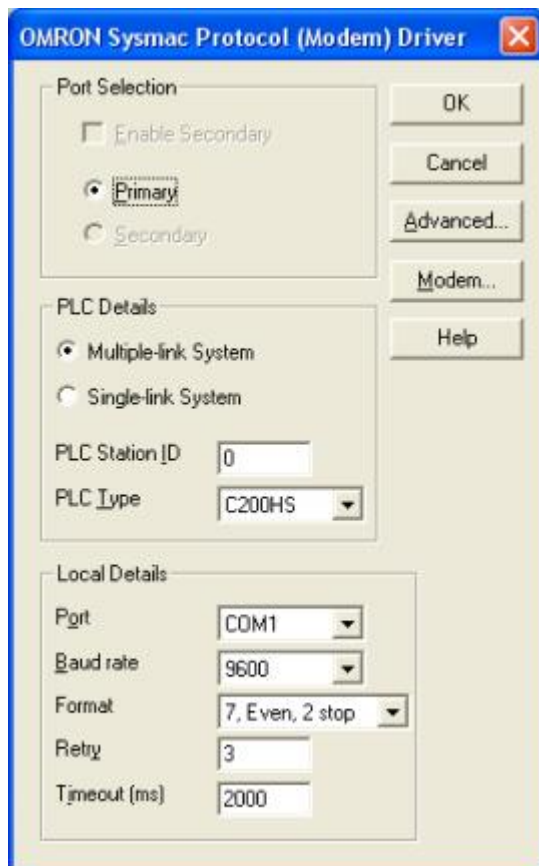
The Modem functions as master to slave dialup only. The drivers polling is now internal , and one must set the poll time for the device (in the modem advanced dialogue) the item scan rate is ignored, all scanitems will be retrieved once per dialup period.

Each device can have its own dialup period.

Notes:

- 1)Due to the polling schedule being on a modem medium, it is recommended to stop all devices before scanning any agent, otherwise the the scan call, will wait till the modem finishes the current device which could be a minute or more.
- 2)It is also recommended to not mix baud rates for different devices I find some modems give problems when training between devices with different baud rates. Even to the point of requiring a hard power down to resolve.

Device configuration



Note: The most commonly used settings are for multi-link system operation with station zero even when on RS232. A baud rate of 9600 is. Re-start your agent server after changing any of these settings for them to take effect.

PLC Details

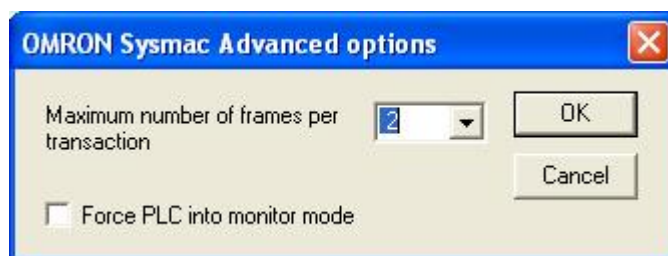
- Multi-link System** Select to enable multiple link protocol on a RS-422 line. If this option is selected, the Link Unit number must be entered in the 'PLC Station ID' field.
- Single-link System** Select to disable multiple link (for point to point communication).
- PLC Station ID** Host Link unit number for multi-link (multidrop) connections
- PLC Type** Applicable SYSMAC PLC types. The PLC type selected will apply to the primary and secondary port if available. This selection currently has no affect on communications.

Local Details

- Port** The available serial ports on the machine.
- Baud Rate** The speed of data transmission. Any of the following Baud Rates may be specified 300, 600, 1200, 2400, 4800, 9600, 19200 or 38400.
- Format** The transmission format used for the port.
- Retry** The number of times that Adroit will try to gain communication with a PLC before failing a transaction.
- Time-out** The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Advanced dialog

The settings visible in the advanced dialog should only be modified to optimise configuration.

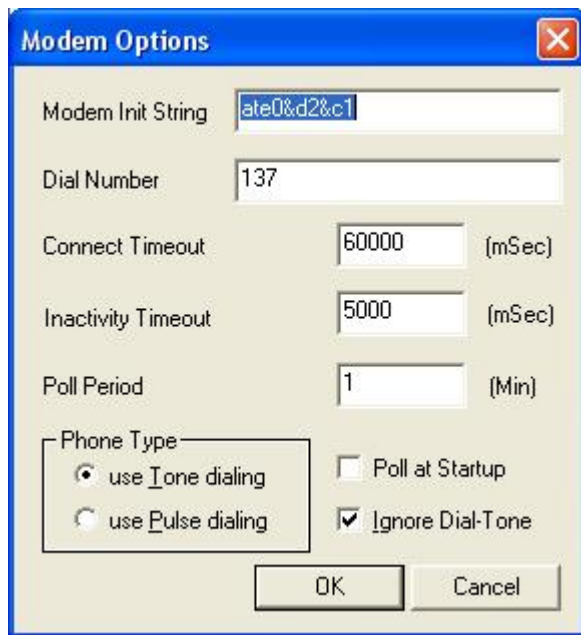


- Maximum number of frames per transaction** The number of frames that Adroit will grow transactions to. A larger number reduces the number of poll records used, but will reduce performance if the scan points are widely scattered.
- Force PLC into monitor mode** Allow the Driver to put the PLC into monitor mode on startup as well as whenever the PLC mode is changed to run. If at any stage the driver attempts to write to PLC memory and fails because the PLC is in run mode this option will allow the driver to set the PLC mode while running. The driver will fail to write values to PLC memory if it is in run mode.

The settings in the advanced dialog will be applied to primary and secondary configurations.

Modem Options

The settings visible in the Modem Options dialogue.



- Modem init string** The initialization string sent to the modem whenever a connect is required. See annexure for additional information.
- Dial Number** Supply the outstation telephone number here.
- Connect timeout** Suggested time at least (30000)30 seconds.
The maximum time to wait for a successful modem connection train. Please note : the driver will attempt to dial 3 times before continuing with the next device.
- Inactivity timeout** (0 Disables)
The maximum time to wait after a successful write. This is a time to wait for any additional writes to the same device, if a poll comes due during this time it will get done without the need for a dial etc.
- Poll Period** (0 Disables – thus will be manual via AD0 admin tag.(see POLL NOW functionality below)
The drivers polling is now internal , and one must set this poll period for the device any item scan rates are ignored, all scanitems will be retrieved once per dialup period.
- Poll at startup** Default = TRUE, If FALSE no initial startup poll will be executed upon agent server startup, only after the first poll cycle, or if manually done.

- Phone type** Select Pulse or Tone exchange.
- Ignore Dial Tone** Will set the modem to ATX3 (x2 Hayes), ignore dialtone, else the modem will not dial till dialtone is established.

Wiring

RS232C from Adroit Master to the Master modem. Usually one will use the cable supplied with the modem. DTE->DCE. The cable from the Outstation modem to the plc will probably be DCE->DTE aswell, depending on the orientation of the PLC (Usually DTE)

(Pc)DTE->(Modem)DCE x (Modem)DCE ->(Pc)DTE

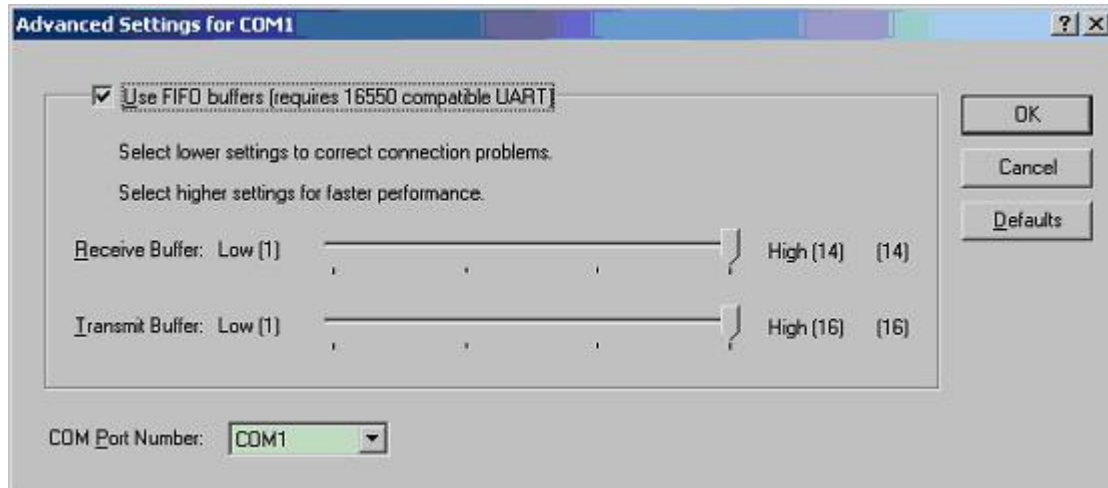
Please note DTE to DTE requires crossover ie tx to rx etc , Where as DTE to DCE uses straight connection tx to tx etc. the crossover then happens unseen between modems.

CPU Comm port 9 pin Male		PC 9 pin Female
1	_____	1
2	_____	2
3	_____	3
4	_____	4
5	_____	5
6	_____	6
7	_____	7
8	_____	8

PC COM Ports – Hardware Setup

If communications appears inconsistent or unreliable, the serial port settings may need some adjusting. Reducing the serial port RX FIFO buffer to 1 (or minimum) may help. If the computer does not have a buffered UART, this setting may not help, in such a case, set the PLC for a lower baud rate.

For Windows 2000 onwards, Right-click on “My Computer”, select manage, then click on device manager. Choose Com & Lpt ports and select the COM port (ie COM1 if you are using COM1). Under the Port settings choose advanced and push the Rx buffer slider to the left most position (ie minimum or 1). Apply changes and the OMRON system should now reply correctly.



Supported addresses

OMRON SYSMAC PLC data addresses supported by Adroit. (* means see note below)

Device	Prefix	Bits	Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STRING
Internal Relay	IR	1*	000.00	8191.15	Bit	IR 000	IR 000	IR 0000	
		16	000	8191	BCD16		IR 000 B	IR 0000 B	
		16	000	8191	Unsigned Binary (Word)		IR 000 W	IR 0000 W	
		16	000	8191	Signed Binary (Integer)		IR 000 I	IR 0000 I	
		32	000	8190	Unsigned Binary (Double Word)		IR 000 D	IR 0000 D	
		32	00	8190	BCD32		IR 000 N	IR 000 N	
		32	000	4095	Signed Binary (Long)		IR 000 L	IR 0000 L	
Holding Relay	HR	1*	00.00	1599.15	Bit	HR 000	HR 000	HR 0000	
		16	00	1599	BCD16		HR 000 B	HR 00 B	
		16	00	1599	Unsigned Binary (Word)		HR 000 W	HR 00 W	
		16	00	1599	Signed Binary (Integer)		HR 000 I	HR 0000 I	
		32	00	1598	Unsigned Binary (Double Word)		HR 000 D	HR 000 D	
		32	00	1598	BCD32		HR 000 N	HR 000 N	
		32	00	1598	Signed Binary (Long)		HR 000 L	HR 000 L	
Auxiliary Relay	AR	1*	00.00	959.15	Bit	AR 000	AR 000	AR 000	
		16	00	959	BCD16		AR 000 B	AR 000 B	
		16	00	959	Unsigned Binary (Word)		AR 000 W	AR 000 W	
		16	00	959	Signed Binary (Integer)		AR 000 I	AR 000 I	
		32	00	958	Unsigned Binary (Double Word)		AR 000 D	AR 000 D	
		32	00	958	BCD32		AR 000 N	AR 000 N	
		32	00	958	Signed Binary (Long)		AR 000 L	AR 000 L	
Link Relay	LR	1*	00.00	1023.15	Bit	LR 000	LR 000	LR 000	
		16	00	1023	BCD16		LR 000 B	LR 000 B	
		16	00	1023	Unsigned Binary (Word)		LR 000 W	LR 000 W	
		16	00	1023	Signed Binary (Integer)		LR 000 I	LR 000 I	
		32	00	1022	Unsigned Binary (Double Word)		LR 000 D	LR 000 D	
		32	00	1022	Signed Binary (Long)		LR 000 L	LR 000 L	
Timer/ counters (preset)	TCP	16	000	2047	BCD16		TCP 000 B		
		16	000	2047	Unsigned Binary (Word)		TCP 000 W		
		16	000	2047	Signed Binary (Integer)		TCP 000 I		
Timer/ counters (current)	TCC	16	000	2047	BCD16		TCC 000 B		
		16	000	2047	Unsigned Binary (Word)		TCC 000 W		
		16	000	2047	Signed Binary (Integer)		TCC 000 I		
Data Memory	DM	1	0000.00	9999.15	Bit	DM 000			
		16	0000	9999	BCD16		DM 000 B	DM 000 B	
		16	0000	9999	Unsigned Binary (Word)		DM 000 W	DM 000 W	
		16	0000	9999	Signed Binary (Integer)		DM 000 I	DM 000 I	
		32	0000	9998	Unsigned Binary (Double Word)		DM 000 D	DM 000 D	
		32	0000	9998	Signed Binary (Long)		DM 000 L	DM 000 L	
		32	0000	9998	Float			DM 000 F	
		32	0000	9998	IEEE754Float			DM 000 R	
Admin Area	AD	1			Bit				
		16			BCD				
		16			Unsigned Binary (Word)				
		16	0	0	Signed Binary (Integer)		AD 0		
		32			Unsigned Binary (Double Word)				
		32			Signed Binary (Long)				
		32			Float				
		32			IEEE754Float				

- * Bit writes are done using a Read-Mask-Write algorithm to achieve an effective bit write. The Read-Mask-Write algorithm reads a word from PLC memory, masks in the bit to be written and immediately writes the entire word to PLC memory. Therefore if the PLC program modifies the bits between the *Adroit* drivers read and subsequent write operations, the changes made by the PLC program will be lost.

Other OMRON-Protocol PLCs

KEYENCE

Settings to use are 9600, 8 data and even parity. The protocol is the same.

Process Plus

Process Plus modules are supported, an instruction leaflet is provided with the modules on how to configure devices to operate with the I/O modules.

Message protocol

The message format for multiple and single connections must be seen as a wrapper around the messages detailed below. (x10) is used to denote numbers in base 10, (x16) is used to denote numbers in HEX.

Message format for multiple connection protocol:

The message format below should be applied if multiple link selection is made in the Driver setup dialog.

		Unit no.		Header and text (see below for contents)		FCS		Terminator	
@		0	0	Header	Text	X	X	*	<CR>
FCS calculation range									

Message format for single link protocol:

Header and text (see below for contents)		FCS		Terminator	
Header	Text	X	X	*	<CR>
FCS calculation range					

IR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	R	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	R	0	0	000A	000B

IR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	R	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
R	R	0	0

HR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	H	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	H	0	0	000A	000B

HR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	H	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	H	0	0

AR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	J	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	J	0	0	000A	000B

AR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	J	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	J	0	0

LR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	L	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	L	0	0	000A	000B

LR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	L	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	L	0	0

TC READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	C	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	C	0	0	000A	000B

TC WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	C	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	C	0	0

DM AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	D	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	D	0	0	000A	000B

DM AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	D	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	D	0	0

STATUS WRITE:

Used to set the PLC back to monitor mode if configured to do so. (see advanced... configuration dialog box)

Send Telegram:

Header		Mode (x16)	
S	C	0	2

Reply Telegram:

Header		Response code (x16)	
S	C	0	0

Performance

This driver supports the standard performance monitor counters described in the user manual.

Advanced registry settings

If the device was created with a driver revision less than 95 you will have to manually add the following in :

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit
Technologies\Adroit\ProtocolDrivers\OMRSYSMM\<device name>\Primary

Add (DWORD) FloatOrder = 1 if rev > 94. To do a word swap on the 32bit floating types.

FloatOrder = 2. To do a word swap then a byte swap on the 32bit floating types.

FloatOrder = 0. (or not configured) No swapping.

Add (DWORD) LongOrder = 1 if rev > 94. To do a word swap on the 32bit floating types.

LongOrder = 2. To do a word swap then a byte swap on the 32bit floating types.

LongOrder = 0. (or not configured) No swapping.

For the above only the primary keys are use by both primary and secondary configuration(s).

Troubleshooting

If at first there is no communication, try setting the system to “multi-link” mode with station address 0 (zero).

Simplified cable diagram with fewer conductors (only use this diagram if you have to)

CPU Comm port 9 pin Male		PC 25 pin Female	PC 9 pin Male
2	————	3	2
3	————	2	3
7	] [	4	7
8		5	8
5	————	7	5

If the floats look very strange (very small or very big), try the new IEEE754 float “R” type , and/or the FloatOrder registry key changes.

If the longs look very strange (very small or very big), try the LongOrder registry key changes.

Administrative functions

Scan any INTEGER type agent to scan address “AD 0”, Make sure it is output enabled.

When ever any value is written to this tag the driver will be prompted to do a certain task... see below

Scan 1 agent for every device, this will use scan licence point.

1 - Manual scanning (poll now)

In order to get the driver to dialup and poll an outstation manually

When ever the value of “1” is written to this tag the driver will be prompted to (dial if not already connected) and poll the associated outstation.

2 – Continuous polling

In order to get the driver to dialup and poll an outstation continuously without stopping until stopped manually via 3 below

Write a value of “2” is to this tag the driver will be prompted to (dial if not already connected) and poll the associated outstation continuously until stopped via 3 below.

NB – Any Control to any other physical device (different number) will stop continuous polling , and do the write , continuous polling will not resume.

3 – Stop Continuous polling

In order to get the driver to stop continuously polling a device , write a “3” into this tag.

Appendix A – How to troubleshoot and setup modems.

This discussion describes landline dialup modem support , but gsm modems are also compatible to the basic command sets so there is no reason for them not to work either.

Where applicable I will refer to the Hayes(x.2) command set as well as the Duxbury command set.

(*h) – X2 Hayes only (e.g US robotics)

(*d) – Duxbury only.

Nothing means identical for both command sets.

1st steps:

Use hyperterminal to establish initial communications with the modems , make sure you can communicate with the modem by creating a hyper terminal session manually to the appropriate com port. Type in any command and check whether the modem acknowledges it. Try AT(return) you should see OK. [ATZ, ATI4(*h) or AT&V(*d) for current config info etc)

If no ack is found try a command that will definitely return something ATI4(*h) AT&V(*d).

If no acks are sent but the about info does get sent check result codes are being sent Q = 0. And the codes must be returned as text V=1.

AT&F will reset the modem factory defaults.

Make sure Carrier detect is enabled &C1. *NB – for driver hang up ability.

Make sure DTR control is enabled &D2. *NB – for driver hang up ability.

If you want the modem to auto-answer make sure S0>0 (i.e. S0=2 for 2 rings then answer)

Make sure Echo is off when you are finished testing E=0. *NB – for driver logic.

Make sure B0(*h). Data baud rate will follow connection train rate.

AT&W(*h) will save the profile , older modems have 2 profiles AT&W0 , AT&W1...

For Remote modem

Make sure the following is set : S0=2, The PLC probably won't answer manually.

And &D0 as the PLC again probably does not control the DTR line and thus must be disabled/ignored! Else the modem will not train properly

Recommended ATE0 s0=0 &D0 &C1&W done manually via terminal! Make sure the save went through by turning off the modem and then doing a ATI4(*h) to check.

For Master modem

S0 = 0, You don't want the modem to auto answer.

&D2 Spectr Driver must have control of the DTR line to do proper hangup! Make sure these control line are wired into the DTE->DCE as specified above.

&C1 Driver must have be able to see if Carrier has been established.

E0 Echo must be off! The driver cannot connect if echo is on

Recommended connection string ATE0s0=0&D2&C1&W some modems don't like the &W it them must be done manually via terminal!

[illegible]

Revisions

Date	Revision
21-Sep-04	Rev96 - Modem based OMRON Sysmac driver first release, converted from rev94 of OMREON Sysmac serial based driver.
22-Sep-04	Rev97 – Poll now functionality added and some code stability. Mark bad if comms fail , some fixes!
28-Sep-04	Rev98 – Float/Long order registry keys added, and updated instructions string to show IEEE754 float “R” type.
10-Oct-06	Rev100 – Updated driver to match omrsysmc, including BCD types